# Project Development Phase

| Date | 2 March 2026 |
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| Team Members Name | Aditi Parashar , Adarsh Singh, Aditya |
| Project Name | Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau |

## Preprocessing Steps:

The preprocessing phase was carried out on the dataset "Transformed_Housing_Data2.csv" to prepare it for building the House Data Analysis Dashboard.

1. **Understanding the Housing Dataset**

- Data types of all columns were verified.
- Sale price distribution was examined to identify extreme values.
- Null values and zero values (especially in renovation year) were checked.

This step ensured clarity about the structure and quality of housing data.

**2. Cleaning and Formatting for Housing Analysis**

To ensure accurate housing insights:

- Sale Price was verified as numeric for calculating **Average Sale Price** .
- Area-related columns were formatted correctly for aggregation.
- Year Built and Year Renovated were validated as numeric fields.
- Records with incorrect or inconsistent values were reviewed.

This was necessary for generating reliable KPIs and visualizations.

**3. Structuring Data for Dashboard Goals**

The dataset was structured based on dashboard requirements:

- Created price ranges for **Market Concentration Analysis**
- Prepared data for:
    - Packed Bubble Chart (Price Distribution)
    - House Age Distribution
    - Renovation Status Analysis
    - Feature Comparison (Bedrooms, Bathrooms, Floors)

**5. Creation of Calculated Field**: Age Since Last Renovation

Condition :- IF [Ever Renovated Yes] > 0 THEN "YES" ELSE "NO" END

- To analyze property condition, a new calculated column was created:

  Age Since Last Renovation

- This field helps measure how recently a property was renovated.

**6. Validation of Key Metrics**

- Average Sale Price was cross-verified.
- Total house count was checked.
- Aggregated values used in packed bubble and bar charts were validated.
- Calculated fields were tested for logical correctness.

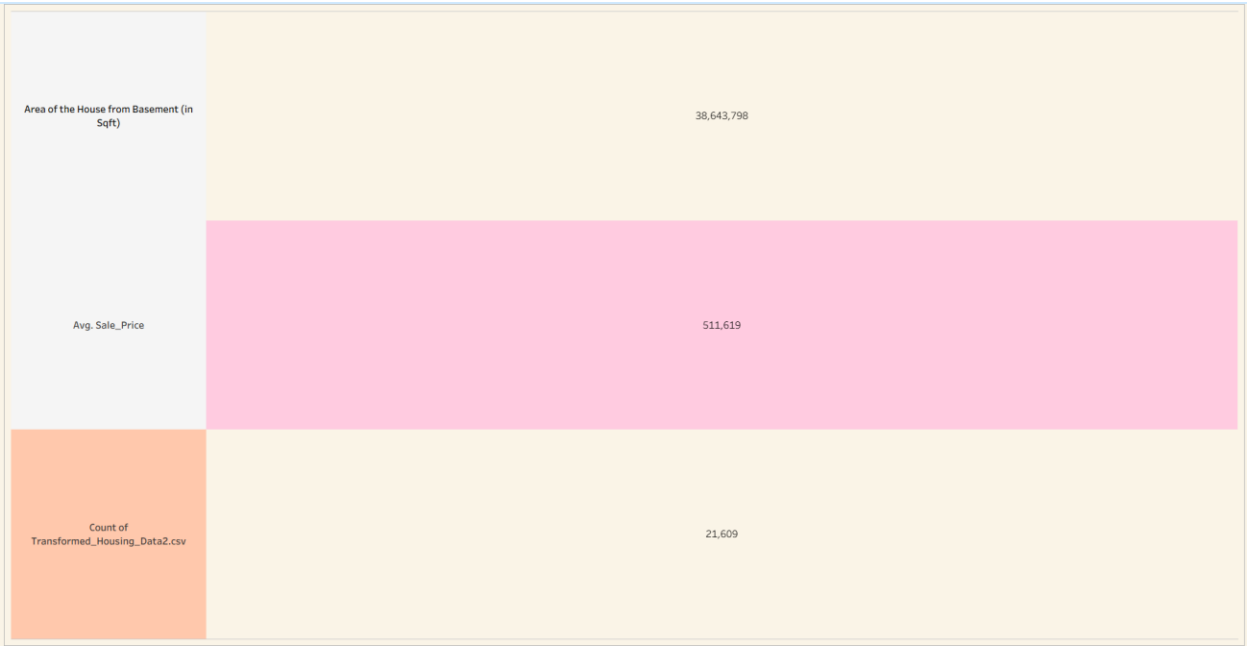
This ensured the insights presented in the dashboard are accurate and reliable.

# Visualizations and Business Questions

**1. KPI Summary Visualization**

(Basement Area, Avg Sale Price, House Count)

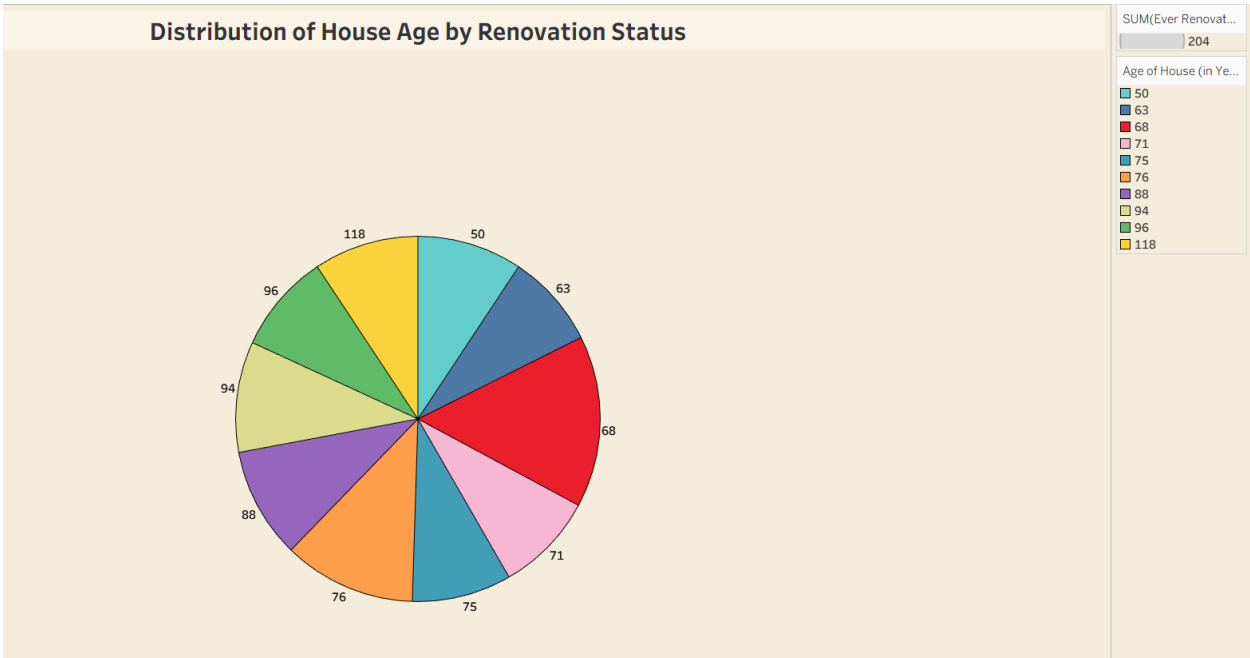
**Business Question:**
How does overall house size (especially basement area) influence the average selling price in the housing market?




## 2. Distribution of House Age by Renovation Status

**Business Question:**

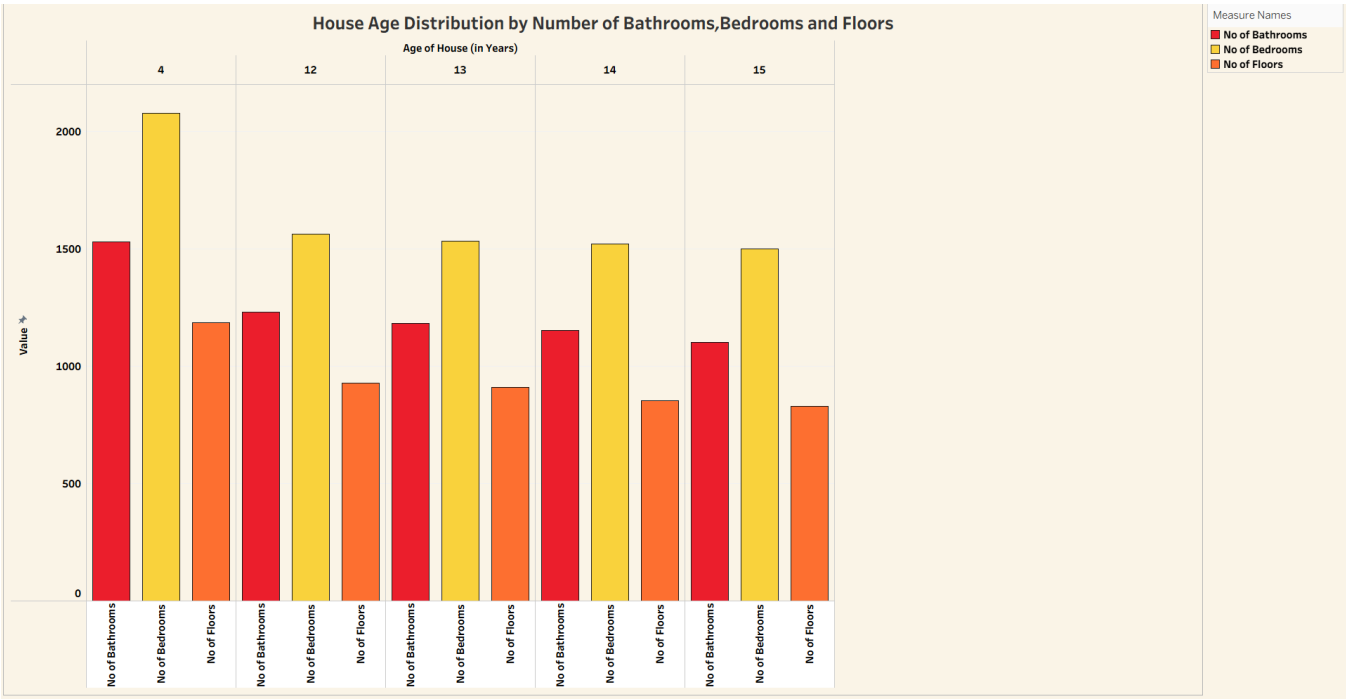
Which age groups of houses are most commonly renovated, and should investors target older homes for renovation opportunities



### 3. House Age Distribution by Number of Bathrooms, Bedrooms and Floors

**Business Question:**

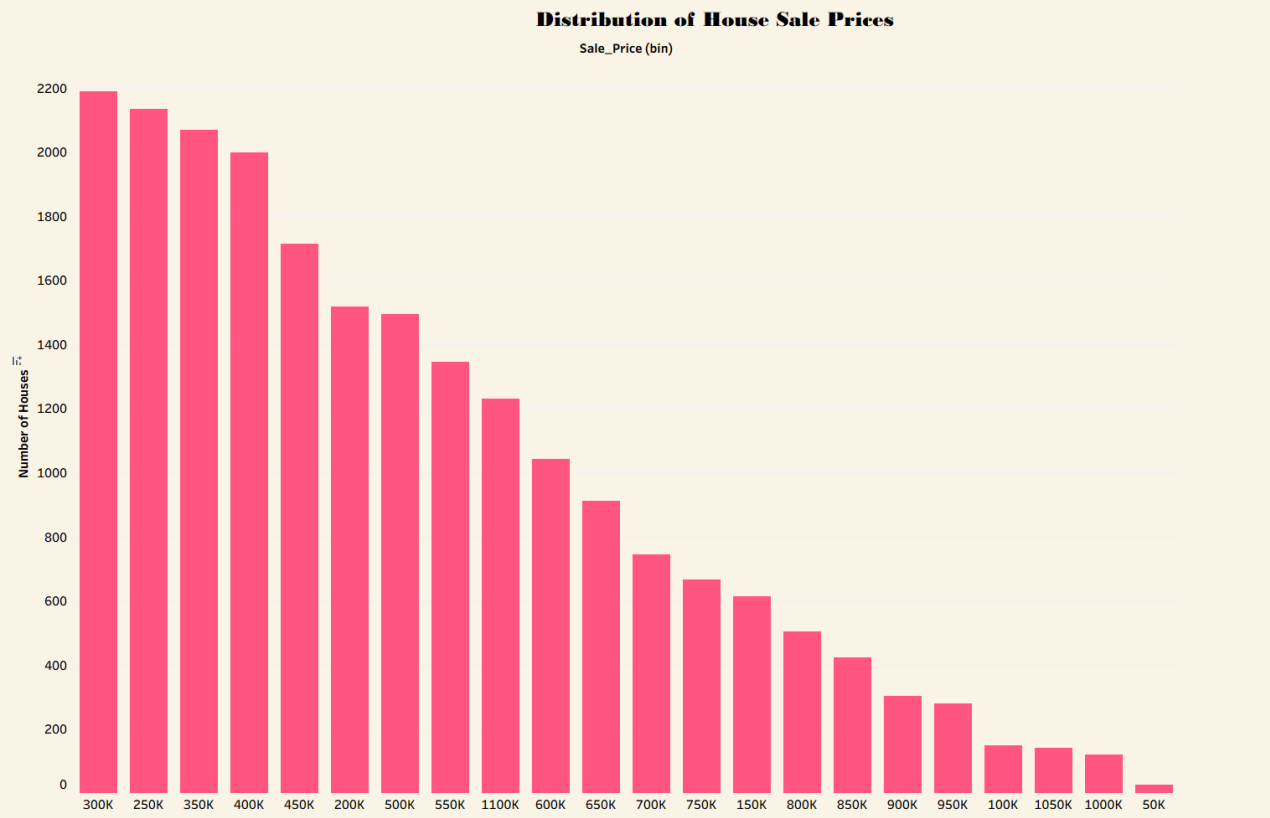
How do bedrooms, bathrooms, and floors differ across house ages, and which age group offers the most features?




### 4. Distribution of House Sale Prices

**Business Question:**

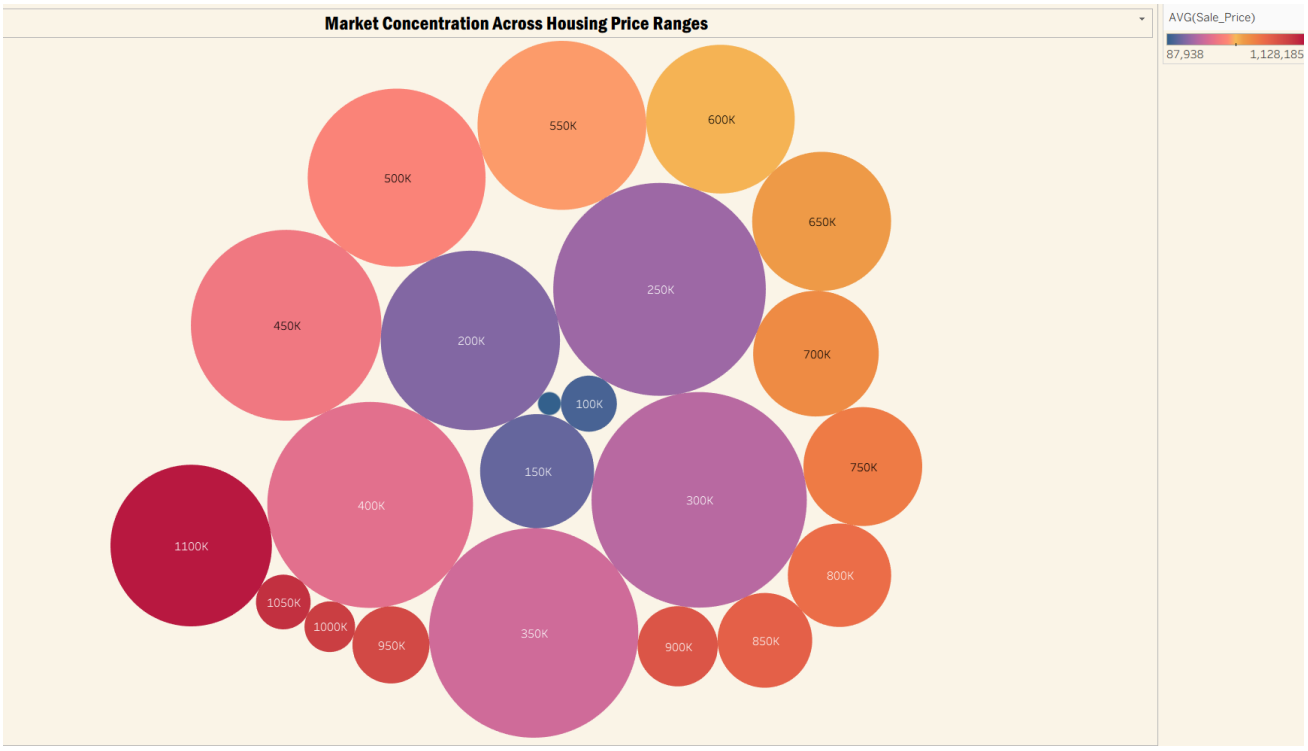
What price range dominates the housing market, and where is the highest concentration of house sales?




## 5. Market Concentration Across Housing Prices Ranges

**Business Question:**

Which housing price ranges have the highest market concentration, and where should agents focus their efforts?


Market Concentration Across Housing Price Ranges
AVG(Sale_Price)
87,938
1,128,185
550K
600K
500K
650K
250K
450K
200K
700K
100K
150K
750K
400K
300K
1100K
800K
1050K
1000K
950K
350K
900K
850K